



LECTURE 5

DEEP REPRESENTATION LEARNING II

Learning beyond Memorization

PRESS – OR SPACE. →

In this lecture

- Neural networks and learned representations
- Supervised training and optimization
- Learning useful intermediate features directly from data
- Convolutional neural networks
- Geospatial and remote-sensing examples

Why the shape of the loss landscape matters, and how regularization keeps models from overfitting it.

Loss Surfaces and Regularization

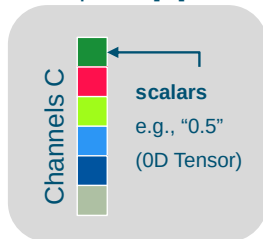
Data and Experience



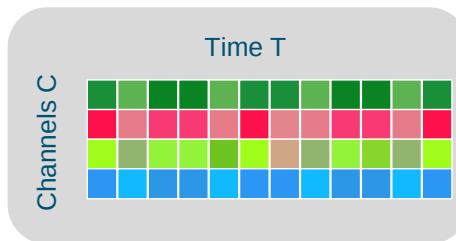
Data is the window through which a learner experiences the world.

Data as blocks of numbers - E.g., Tensors

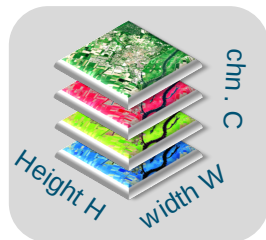
1D Tensor: Vector
e.g., pixels or
class probs [C]



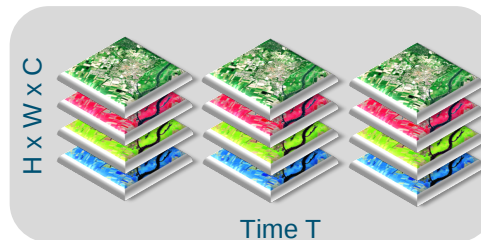
2D Tensor: Matrix
Time Series [T x C]



3D Tensors
Images [H x W x C]



4D Tensors
Img. Time Series [T x H x W x C]



Data samples from a distribution